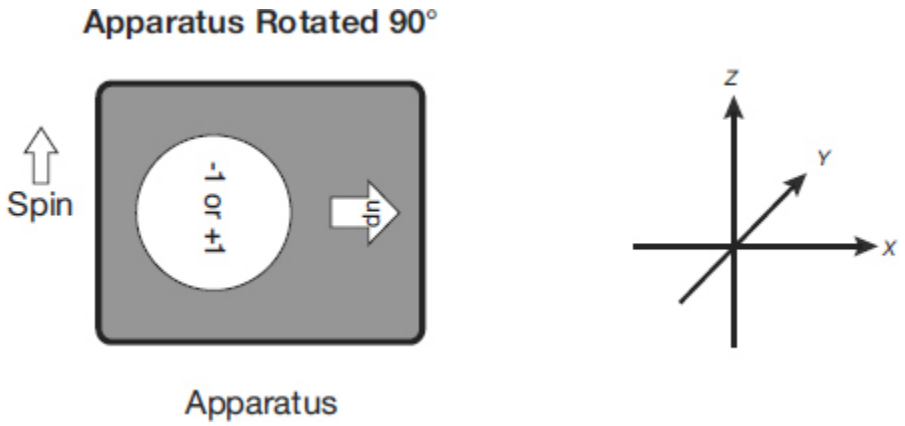


Figure 1.3: The apparatus rotated by 90°. A new measurement results in $\sigma_x = -1$ with 50 percent probability.

So far, there is still no difference between classical physics and quantum physics. The difference only becomes apparent when we rotate the apparatus through an arbitrary angle, say $\frac{\pi}{2}$ radians (90 degrees). The apparatus begins in the upright position (with the up-arrow along the z axis). A spin is prepared with $\sigma = +1$. Next, rotate $\mathcal{A}$ so that the up-arrow points along the x axis (Fig. 1.3), and then make a measurement of what is presumably the x component of the spin, σ_x.

If in fact σ really represents the component of a vector along the up-arrow, one would expect to get zero. Why? Initially, we confirmed that σ was directed along the z axis, suggesting that its component along x must be zero. But we get a surprise when we measure σ_x: Instead of giving $\sigma_x = 0$, the apparatus gives either $\sigma_x = +1$ or $\sigma_x = -1$. $\mathcal{A}$ is very stubborn—no matter which way it is oriented, it refuses to give any answer other than $\sigma = \pm 1$. If the spin really is a vector, it is a very peculiar one indeed.

Nevertheless, we do find something interesting. Suppose we repeat the operation many times, each time following the same procedure, that is:

- Beginning with $\mathcal{A}$ along the z axis, prepare $\sigma = +1$.
- Rotate the apparatus so that it is oriented along the x axis.